

1. Administrative & Study Identification

Full Study Title: An Open-Label, Multicenter Phase 2 Basket Study to Evaluate the Antitumor Activity and Safety of Lenvatinib in Children, Adolescents, and Young Adults With Relapsed or Refractory Solid Malignancies **Short Title/Acronym:** A Study of Lenvatinib (MK-7902) in Pediatric Participants With Relapsed or Refractory Solid Malignancies (MK-7902-013/E7080) **Protocol Number:** MK-7902-013 (E7080-G000-231) **NCT Number:** NCT04447755 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Lenvatinib (MK-7902 / E7080 / LENVIMA®) | **ATC Code:** L01EX08 **Phase of Development:** Phase 2 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the antitumor activity and safety of lenvatinib (MK-7902/E7080) in children, adolescents, and young adults with relapsed or refractory solid malignancies. **Secondary Objectives:** To evaluate Overall Objective Response Rate (ORR), Progression-Free Survival (PFS), Best Overall Response (BOR), Duration of Response (DOR), Overall Survival (OS), and the palatability/acceptability of the lenvatinib suspension formulation.

2.2 Background & Rationale There remains a critical, unmet medical need for effective therapies in pediatric and young adult patients with relapsed or refractory solid tumors. While lenvatinib (a multiple receptor tyrosine kinase inhibitor) has proven efficacy in adult cancers, this basket study investigates its single-agent activity and safety profile across specific pediatric cohorts—including Ewing sarcoma (EWS), rhabdomyosarcoma (RMS), high-grade glioma (HGG), diffuse midline glioma, medulloblastoma, ependymoma, and other solid tumors (excluding osteosarcoma)—to expand therapeutic options for these vulnerable populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Basket Study) **Blinding:** Open-label **Randomization:** N/A **Trial Status:** COMPLETED

3.2 Planned Enrollment & Duration Target \$N\$: 127 participants (Actual Enrollment) **Number of Sites:** Multicenter, global

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Children, adolescents, and young adults with histologically or cytologically documented relapsed or refractory pediatric solid malignancy. 2. Measurable disease as defined by RECIST 1.1 or Response Assessment in Neuro-Oncology (RANO) for High Grade Glioma (HGG). 3. Performance status: Lansky Play Score ≥ 50 (for patients ≤ 16 years of age) or Karnofsky performance status (KPS) ≥ 50 (for patients >16 years of age). Neurologic deficits in participants with primary central nervous system (CNS) tumors must be stable for at least 7 days prior to enrollment. 4. Adequate organ, cardiac, and neurologic function. 5. Adequate blood pressure (BP) control with or without antihypertensive medications, and no clinical evidence of nephrotic syndrome. 6. Recovery to Common Terminology Criteria for Adverse Events (CTCAE v5.0) Grade ≤ 1 (except for alopecia, ototoxicity, and Grade ≤ 2 peripheral neuropathy) from the acute toxic effects of all prior anticancer therapy. 7. Agreement to use approved contraception during the treatment period and for required windows post-treatment (at least 7 days for males, 30 days for females of childbearing potential).

4.2 Exclusion Criteria 1. Diagnosis of osteosarcoma (evaluated in a separate trial program). 2. Major surgery within 3 weeks prior to Cycle 1 Day 1 (C1D1). 3. Gastrointestinal (GI) bleeding or active hemoptysis within 21 days prior to enrollment, or preexisting $\geq$ Grade 3 GI or non-GI fistula, GI malabsorption, or GI anastomosis. 4. CNS tumors with a history of symptomatic tumor hemorrhage, evidence of new intracranial hemorrhage of more than punctate size within 28 days prior to enrollment, or untreated CNS metastases (exception: primary CNS tumors and leptomeningeal disease). 5. Radiographic evidence of encasement or invasion of a major blood vessel or of intratumoral cavitation. 6. Any active infection requiring systemic therapy, known HIV positivity, or known active viral hepatitis (B or C). 7. Concurrent participation or receipt of an investigational agent/device within 4 weeks of allocation. 8. Clinically significant cardiovascular disease within 6 months (e.g., NYHA Class III or IV congestive heart failure, unstable angina, myocardial infarction, cerebral vascular accident, or cardiac arrhythmia with hemodynamic instability). 9. Non-healing wound, tumor ulceration, unhealed/incompletely healed fracture, or a compound bone fracture at the time of enrollment. 10. Known psychiatric, substance abuse disorders, or known hypersensitivity to any component of lenvatinib.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Lenvatinib 14 mg/m² once daily (QD). **Route of Administration:** Oral (Capsules or via an oral suspension formulated with water or apple juice). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptoms/co-morbidities, including antiemetics and antihypertensive medications. **Prohibited:** Concurrent investigational agents or other systemic anticancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Demographics, KPS/Lansky score, Tumor Imaging (Baseline), Organ/Cardiac Function Labs, Hepatitis/HIV Serology (if applicable)
Baseline	Day 0	Enrolment, First Dose of Lenvatinib, Formulation Palatability Assessment
Interim	Every 4 to 8 Weeks	Safety Labs, Efficacy Assessment (RECIST 1.1 / RANO), BP Monitoring, AE review
End of Study	Variable	Treatment Discontinuation Visit, Final Vital Signs, Exit Imaging, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR) At Week 16 by Investigator Assessment. Defined as the percentage of participants with a confirmed complete response (CR: disappearance of all target lesions) or partial response (PR). **Measurement Method:** * **Solid Tumors:** Assessed by the investigator using RECIST 1.1 (PR defined as at least a 30% decrease in the sum of diameters [SOD] of target lesions). * **HGG Only:** Assessed using RANO criteria where overall response is based on radiographic response (PR defined as sum of products of diameters [SPD] decreased by $\geq$ 50% from baseline) and clinical performance status with steroid dose information.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) overall per RECIST 1.1 or RANO criteria. * Progression-Free Survival (PFS) per RECIST 1.1 or RANO criteria. * Best Overall Response (BOR) per RECIST 1.1 or RANO criteria. * Duration of Response (DOR) and Overall Survival (OS). * Palatability and acceptability of the lenvatinib suspension formulation.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection at all visits graded according to the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. Specific monitoring for hypertension, proteinuria, hemorrhage, and hepatic toxicity. **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent safety monitoring committee overseeing pediatric toxicity thresholds.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Analysis Set for all AE/SAE evaluations and palatability assessments. **Sample Size Justification:** Designed as a basket study powered to detect a minimum objective response signal within distinct pediatric tumor cohorts. **Handling of Missing Data:** Missing target lesion measurements are censored at the last evaluable time point for PFS and DOR calculations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in strict accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and applicable pediatric regulatory protections. **Monitoring Plan:** Regular onsite and centralized remote monitoring of site data. **Audit Trail:** Secure eCRF locking mechanisms and rigorous data clarification procedures by the Sponsor.

11. Study Identifiers & Governance

Primary ID: {{MK-7902-013}} **Registry Number:** {{NCT04447755}}
Sponsor/Lead Organization: {{Merck Sharp & Dohme LLC}} **Study Title:** {{A Study of Lenvatinib (MK-7902) in Pediatric Participants With Relapsed or Refractory Solid Malignancies}}
Official Regulatory Title: {{An Open-Label, Multicenter Phase 2 Basket Study to Evaluate the Antitumor Activity and Safety of Lenvatinib in Children, Adolescents, and Young Adults With Relapsed or Refractory Solid Malignancies}}

12. Clinical Framework (Oncology Specific)

Primary Condition: {{Relapsed or Refractory Solid Tumors}}
Diagnosis Threshold: {{Measurable disease per RECIST 1.1 or RANO}} **Medical Methodology:** {{Targeted Tyrosine Kinase Inhibitor Therapy}} **Optimization Target:** {{Objective Response Rate (ORR) and tumor shrinkage}}

13. Study Procedures & Methodology

Management Pathway: {{Lenvatinib Monotherapy (14 mg/m² QD)}}
Education Protocol (HCP): {{Guidance on AE management, particularly hypertension and proteinuria}} **Education Protocol (Patient):** {{Suspension formulation preparation and administration guidelines}} **Quality Audit Mechanism:** {{Investigator assessment of imaging with Sponsor oversight}}

14. Observation & Follow-Up Schedule

Total Duration: {{Treatment until disease progression or unacceptable toxicity}} **Onsite Visit Cadence:** {{Every 4-8 weeks for tumor assessment and clinical evaluation}} **Remote Monitoring Frequency:** {{Continuous safety and adverse event monitoring}} **Checklist Review Interval:** {{Every treatment cycle}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				